JUAN DIEGO DÍAZ AVENDAÑO

Full-Stack Developer | Java, Spring Boot, React, React Native, TypeScript

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PROFESSIONAL SUMMARY

Full-stack developer with 3 years of experience building complete products: REST APIs and microservices with Java and Spring Boot, React and Angular frontends, and React Native and Expo mobile apps, on PostgreSQL and Firebase backends. Delivered production systems for different kinds of organizations, such as a university, a demolition company and a dental practice. Comfortable owning a feature from requirements to deployment. I work with AI coding agents (Claude Code, Codex) inside a process with specs, tests and review. Winner of TechU Montería 2025 (Montería City Hall) and 1st place in the Universidad de Córdoba Systems Engineering Hackathon. Available immediately for full-time remote work.

TECHNICAL SKILLS

Languages: Java, TypeScript, JavaScript, SQL

Backend: Spring Boot, Spring Security, Spring Data JPA, Hibernate, REST APIs, Microservices, JWT, Node.js, Webhooks, Zod

Frontend: React, TypeScript, JavaScript, Angular, HTML, CSS, Tailwind CSS, Vite, React Router, TanStack Query, Axios

Mobile: React Native, Expo, EAS Build, iOS, Android, App Store publishing, push notifications, RevenueCat

Databases: PostgreSQL, Firestore, Flyway, data modeling, database migrations, optimistic locking

Cloud & DevOps: Docker, Docker Compose, Nginx, Firebase (Auth, Firestore, Cloud Functions, Hosting, App Check), Railway, Vercel, Git, GitHub, CI/CD

Testing: JUnit, Jest, Testing Library, Firebase Emulator tests, Maestro E2E

AI-assisted engineering: Claude Code, Codex, agent harnesses (CLAUDE.md, AGENTS.md, subagents, custom skills), autonomous loops, adversarial review

WORK EXPERIENCE

Full-Stack Developer | Universidad de Córdoba

Montería, Colombia (Remote) | Aug 2025 – Jun 2026

- Built SIGCORP end to end, the Communications Office's request-management web app: REST APIs, assignment, status tracking, role-based access and metrics, with a React and TypeScript frontend.
- In a load test on 2 CPUs and 2 GB of RAM, SIGCORP sustained 200 concurrent users with p95 under 30 ms and 175 new requests per second, about 5 million per 8-hour workday.
- Designed security and persistence with Spring Security (JWT), Spring Data JPA, Flyway and PostgreSQL, with optimistic locking, and deployed with Docker, Docker Compose and Nginx on institutional infrastructure.
- Built backend modules for the university's e-commerce: Wompi gateway with signature-verified webhooks, idempotent processing to avoid duplicate payments, order states, and a Node.js notification microservice over REST.
- Migrated about 200 pages of the university website to WordPress: redesign, content validation, load-time optimization, responsive design and SEO with redirects.

FREELANCE EXPERIENCE

Backend Developer & Team Lead | Sonrisa – Dental Practice Management Software

Remote | Aug 2026 – Oct 2026

- Designed and built the Java 21 / Spring Boot and PostgreSQL backend for patients, appointments, 19-step clinical records with electronic signature, odontogram, treatment plans, payments and finance.
- Implemented Spring Security with JWT and refresh tokens, JPA/Hibernate and Flyway migrations; immutable signed records and an access audit log to comply with Colombian health-data regulations (Ley 1581, RIPS, CIE-10, CUPS).
- Led a team of three: gathered requirements directly with the client, defined the architecture, reviewed pull requests and integrated the React + TypeScript frontend; deployed with Docker on Railway and Vercel.

Full-Stack Developer | SGO – ESCAMVIOS S.A.S.

Remote | May 2026 – Jun 2026

- Replaced an Excel-based transportation workflow with a React, TypeScript and Firebase web application for trips, projects, employees, attendance and client access.
- Migrated and structured 550+ historical trip records; built KPI dashboards, charts, filters, search and pagination.

- Implemented Firestore Security Rules with role-based and project-level data isolation, write validation and emulator-based security tests; first designed the backend in Spring Boot and PostgreSQL, then migrated it to Firebase to fit the client's scale and budget.

Full-Stack Mobile Developer | Biomech – B2B Fitness SaaS
Remote | May 2023 – Jun 2025

- Built a two-sided mobile app (trainers and athletes) with React Native, Expo and TypeScript: clients, routines, progress tracking, weekly streaks, scheduling, bookings and payments. Published on the App Store.
- Designed a serverless Firebase backend with 30+ protected Cloud Functions, Zod validation, Firebase App Check and Firestore/Storage Security Rules, with multi-organization support for locations, trainers and clients.
- Integrated Wompi payments and RevenueCat subscriptions for App Store and Android billing.
- Built my own multi-agent pipeline with audit and adversarial-verification loops; locked down 48 critical modules with characterization tests before refactoring.

PERSONAL PROJECTS

CETI – AI Financial Education Platform
Personal project | Apr 2025 – Sep 2025

- Built a cross-platform mobile and web app with React Native, Expo, TypeScript and Firebase: gamified lessons, streaks, savings goals, expenses, budgets and charts.
- Built a server-side AI workflow on Cloud Functions that turns text and voice requests into structured actions, with validation, user confirmation, usage quotas and local fallbacks.
- Developed it with my own multi-agent harness: 12 agents, 9 skills and a hook that runs the linter after every agent edit.

The queue – Shared activity queue (PWA)
Personal project | Oct 2026 | github.com/juandda0/cola

- PWA built with Angular 22 (signals, standalone components) and Firebase: real-time queue on Firestore, rules that allow exactly two members, Google sign-in, and push notifications and reminders with Cloud Functions.

AWARDS & RECOGNITION

- **Winner – TechU Montería 2025, Montería City Hall (May 2025).** CTO of Unimarket, a university marketplace with catalogs, auctions, raffles and its own digital currency (UNICOIN); one of 5 winning projects out of 72 proposals.
- **1st place – Systems Engineering Hackathon, Universidad de Córdoba (May 2025).** AulaLink, an attendance-tracking system for schools, built in a team of three: Express.js and TypeScript backend, Tailwind CSS web SPA, Flutter mobile app, and SQL and NoSQL databases.
- **Winner – Innovation Fair, Universidad de Córdoba (May 2025).** Unimarket, as CTO of a four-student team from the Innovarium research group.

INTERNATIONAL CONFERENCE PRESENTATIONS

- **“LUMIO: Automated Financial Management and Education” (lead author).** 8th CIIDIES–COPEIN–RIISS International Congress and 10th Emprende Innova UG Congress, Celaya, Mexico (virtual), Oct 2026.
- **“CETI: Financial Education Platform” (co-author).** Same congress, Celaya, Mexico (in person), Oct 2026.
- **“UNIMARKET” (co-author).** 5th International Congress on Technology, Learning and Education (CITAE 2025), REDIIE, Santiago, Chile (hybrid), Jul 2025.

EDUCATION

B.S. in Systems Engineering (in progress, 7th semester) | Universidad de Córdoba
Montería, Colombia | 2023 – Expected 2027

Member of the Innovarium research group (Faculty of Engineering).

LANGUAGES

Spanish – Native | English – C1 (Professional)